

Section 1: Composition and Formation of Natural Gas

1. **Natural gas is primarily composed of which hydrocarbon?**
 - A) Ethane
 - **B) Methane (CH₄)**
 - C) Propane
 - D) Butane
2. **Methane usually makes up what percentage of natural gas?**
 - A) 10-20%
 - B) 50%
 - **C) 70-95%**
 - D) 100%
3. **Natural gas is formed from the remains of ancient:**
 - A) Giant ferns
 - **B) Marine organisms (Plankton and Algae)**
 - C) Dinosaurs
 - D) Volcanic ash
4. **Natural gas is found in which type of rock?**
 - A) Igneous
 - **B) Sedimentary (Porous rock)**
 - C) Metamorphic
 - D) Granite
5. **Which of these is a byproduct often found with natural gas?**
 - A) Gold
 - B) Coal
 - **C) Crude Oil**
 - D) Iron ore

Section 2: Natural Gas in Pakistan (Reserves & History)

6. **In which year was the first major natural gas field discovered in Pakistan?**
 - A) 1947
 - **B) 1952**
 - C) 1960
 - D) 1971
7. **The largest natural gas field in Pakistan is located at:**
 - A) Mari

- **B) Sui**
 - C) Uch
 - D) Khairpur
8. **Sui Gas field is located in which province?**
- A) Punjab
 - B) Sindh
 - **C) Balochistan**
 - D) Khyber Pakhtunkhwa
9. **Which gas field is the second largest in Pakistan?**
- **A) Mari Gas Field**
 - B) Kandhkot
 - C) Dhodak
 - D) Pirkoh
10. **The Mari Gas field is located in which province?**
- A) Balochistan
 - **B) Sindh**
 - C) Punjab
 - D) KPK
11. **Natural gas in Pakistan is often colloquially called "Sui Gas" because:**
- A) It smells like a needle (Sui)
 - **B) It was first discovered in the Sui region**
 - C) It is thin like a thread
 - D) "Sui" is the Urdu word for gas
12. **Which organization is responsible for the exploration of oil and gas in Pakistan?**
- A) WAPDA
 - **B) OGDC (Oil and Gas Development Company)**
 - C) NEPRA
 - D) PTCL
13. **Which province has the highest number of gas reserves in Pakistan?**
- A) Punjab
 - **B) Sindh**
 - C) Balochistan
 - D) KPK

Section 3: Uses and Transportation

14. **CNG stands for:**
- A) Carbonated Natural Gas
 - **B) Compressed Natural Gas**
 - C) Clean Natural Gas
 - D) Cold Natural Gas
15. **In Pakistan, natural gas is widely used in the production of:**
- A) Steel
 - **B) Chemical Fertilizers (Urea)**
 - C) Paper
 - D) Glass only
16. **Natural gas is transported across Pakistan mainly through:**
- A) Trucks
 - B) Railways
 - **C) Pipelines**
 - D) Tankers
17. **Which two companies manage the gas pipeline network in Pakistan?**
- A) PSO and Shell
 - **B) SNGPL and SSGC**
 - C) OGDC and PPL
 - D) K-Electric and WAPDA
18. **The use of natural gas in power plants to generate electricity is called:**
- **A) Thermal Power Generation**
 - B) Hydro Power
 - C) Solar Power
 - D) Nuclear Power
19. **What is added to natural gas to give it a distinct smell for leak detection?**
- A) Oxygen
 - **B) Mercaptan**
 - C) Helium
 - D) Nitrogen

Section 4: Rapid Fire Review (20-100)

20. **Is natural gas renewable? (Ans: No)**

21. **Main use of gas in Pakistani homes? (Ans: Cooking and Heating)**
22. **What is 'Dry Gas'? (Ans: Gas consisting almost entirely of Methane)**
23. **What is 'Wet Gas'? (Ans: Gas containing liquid hydrocarbons like ethane and propane)**
24. **Gas field located in Dera Bugti? (Ans: Sui)**
25. **Largest consumer of gas in Pakistan? (Ans: Power sector/Industry)**
26. **Which gas is used as fuel in cars? (Ans: CNG)**
27. **Is natural gas lighter or heavier than air? (Ans: Lighter)**
28. **Full form of SNGPL? (Ans: Sui Northern Gas Pipelines Limited)**
29. **Full form of SSGC? (Ans: Sui Southern Gas Company)**
30. **Which gas field is in Ghotki? (Ans: Mari)**
31. **Is natural gas a fossil fuel? (Ans: Yes)**
32. **Does natural gas produce ash when burned? (Ans: No)**
33. **Main pollutant released by burning gas? (Ans: Carbon dioxide)**
34. **What is LNG? (Ans: Liquefied Natural Gas)**
35. **Why is gas liquefied? (Ans: For easier transport over long distances/oceans)**
36. **Temperature to liquefy natural gas? (Ans: -162°C)**
37. **Does Pakistan import LNG? (Ans: Yes)**
38. **First LNG terminal in Pakistan? (Ans: Port Qasim)**
39. **Is natural gas colorless? (Ans: Yes)**
40. **Is natural gas odorless in its pure form? (Ans: Yes)**
41. **Raw material for Urea fertilizer? (Ans: Natural Gas)**
42. **Does burning gas cause acid rain? (Ans: Less than coal/oil)**
43. **What is 'Shale Gas'? (Ans: Gas trapped in shale rock formations)**
44. **Does Pakistan have Shale gas potential? (Ans: Yes, very high potential)**
45. **Which sector uses gas for boilers? (Ans: Industrial)**
46. **What is 'Associated Gas'? (Ans: Gas found with oil)**
47. **What is 'Non-associated Gas'? (Ans: Gas found without oil)**

48. **Can natural gas be used for refrigeration? (Ans: Yes, absorption chillers)**
49. **Main component of LPG? (Ans: Propane and Butane)**
50. **Difference between CNG and LPG? (Ans: CNG is Methane; LPG is Propane/Butane)**
51. **Major gas field in KPK? (Ans: Nashpa/Manzalai)**
52. **Discovery year of Mari field? (Ans: 1957)**
53. **Gas field in Badin? (Ans: Various small fields under SSGC)**
54. **What is 'Flaring'? (Ans: Burning off excess gas at oil wells)**
55. **Is flaring good for the environment? (Ans: No, it wastes energy and pollutes)**
56. **Primary use of gas in Cement industry? (Ans: Heating kilns)**
57. **Does gas burning emit Sulfur Dioxide? (Ans: Very negligible amounts)**
58. **Instrument to detect gas leaks? (Ans: Gas detector/Sniffer)**
59. **Can gas be stored underground? (Ans: Yes, in depleted reservoirs)**
60. **What is a 'Gas Well'? (Ans: A hole drilled to extract gas)**
61. **What is 'Gathering Line'? (Ans: Pipeline from well to processing plant)**
62. **Gas field near Kandhkot? (Ans: Kandhkot Gas Field)**
63. **Does Pakistan produce enough gas for its needs? (Ans: No, there is a shortage)**
64. **What is 'Gas Load Shedding'? (Ans: Temporary suspension of gas supply)**
65. **Major cause of gas shortage in winter? (Ans: High demand for domestic heating)**
66. **What is 'Tight Gas'? (Ans: Gas trapped in very hard rock)**
67. **Does natural gas help in reducing 'Carbon Footprint' compared to coal? (Ans: Yes)**
68. **What is 'Casing' in a gas well? (Ans: Steel pipe lining the well)**
69. **Process of removing impurities from gas? (Ans: Gas Processing/Refining)**
70. **Common impurity in gas? (Ans: Hydrogen Sulfide/Carbon Dioxide)**

71. **What is 'Sour Gas'? (Ans: Gas with high Hydrogen Sulfide content)**
72. **What is 'Sweet Gas'? (Ans: Gas with low/no Sulfur content)**
73. **Is Sui gas considered 'Sweet' or 'Sour'? (Ans: Mostly Sweet)**
74. **What is 'Hydrocarbon'? (Ans: Compound of Hydrogen and Carbon)**
75. **Density of Methane vs Air? (Ans: 0.55 to 1.0)**
76. **Energy content of natural gas is measured in? (Ans: BTU - British Thermal Units)**
77. **Which province consumes most gas? (Ans: Punjab)**
78. **Gas field in Lasbela? (Ans: Uch Gas Field)**
79. **What is 'Town Border Station' (TBS)? (Ans: Where gas pressure is reduced for city use)**
80. **Does gas pressure decrease over time in a field? (Ans: Yes)**
81. **What are 'Compressor Stations'? (Ans: Stations to boost pressure in pipelines)**
82. **Is natural gas used in the Textile industry? (Ans: Yes, for captive power and processing)**
83. **Can natural gas be made from waste? (Ans: Yes, as Biogas/Biomethane)**
84. **What is the main pipeline project from Iran? (Ans: IP Pipeline)**
85. **What is TAPI? (Ans: Turkmenistan-Afghanistan-Pakistan-India Pipeline)**
86. **Is natural gas explosive? (Ans: Yes, when mixed with air in certain ratios)**
87. **Lower explosive limit (LEL) of Methane? (Ans: 5%)**
88. **Upper explosive limit (UEL) of Methane? (Ans: 15%)**
89. **Does water vapor need to be removed from gas? (Ans: Yes, to prevent corrosion/freezing)**
90. **What are 'Condensates'? (Ans: Liquid hydrocarbons that drop out of gas)**
91. **Common unit for gas volume? (Ans: Cubic Feet / Cubic Meters)**
92. **What is MCF? (Ans: Thousand Cubic Feet)**
93. **What is MMCFD? (Ans: Million Cubic Feet per Day)**

94. Which company discovered Sui gas? (Ans: **Pakistan Petroleum Limited - PPL**)
95. Does natural gas contain Oxygen? (Ans: **Usually no**)
96. Can natural gas be used to make Plastic? (Ans: **Yes**)
97. Major gas field in Jhelum/Punjab? (Ans: **Adhi (Oil & Gas)**)
98. Is natural gas visible? (Ans: **No**)
99. Does natural gas help in 'Industrialization'? (Ans: **Yes, it is a key fuel**)
100. The future of gas in Pakistan depends on? (Ans: **New explorations and LNG imports**)